10. Fluorine exists naturally as the fluoride ion. It is found in soil, water, foods and several minerals, such as fluorapatite and fluorite.

Fluoride ion concentration in seawater averages 1.3 ppm (parts per million). In fresh water, the natural range is typically between 0.01 and 0.3 ppm. In some parts of the world, fresh water contains fluoride ion levels which are dangerous and can lead to health problems.

In the early 1930s, scientists found that people who were brought up in areas with naturally fluoridated water had up to two-thirds fewer cavities compared to those who lived in areas where the water was not fluoridated. Several studies since then have repeatedly shown that when fluoride is added to people's drinking water in areas where natural levels are low, tooth decay decreases.

However, many European countries which do not fluoridate their water do not have a higher incidence of dental decay than countries which do so. It was also found that in Germany and Finland, decay rates either remained stable or continued in their downward trend after they stopped adding fluoride to their drinking water.

Figure 1 shows data about the effect of fluoridation of drinking water on the mean number of decayed, missing and filled teeth (DMFT) and the amount of fluorosis seen.

Figure 2 shows the change in mean DMFT in three regions of Australia over a four year period.

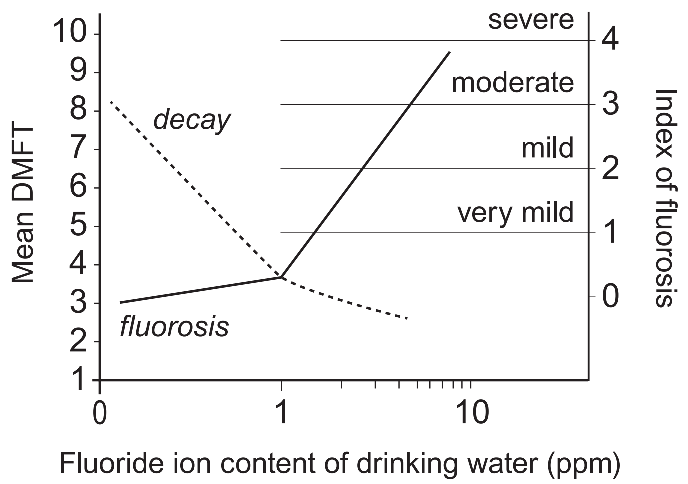

Figure 1

Examiner only

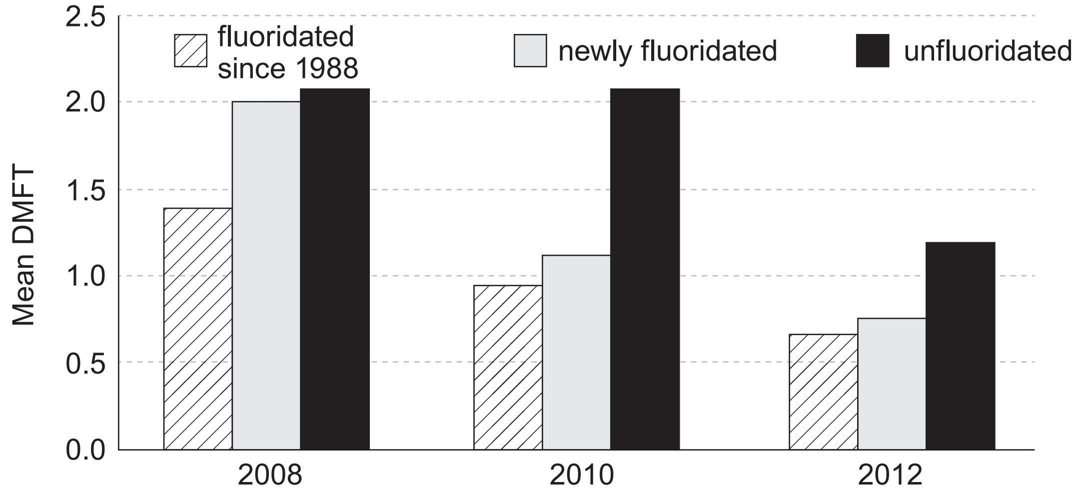

Figure 2

(a) Describe the effects of adding varying concentrations of fluoride ions to drinking water. [3]

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(b) Tick (✓) the statement below that best describes why Germany stopped fluoridating its water supplies. [1]

Ten years after stopping adding fluoride there was no increase in tooth decay ☐

They found that adding fluoride caused fluorosis ☐

Natural water supplies already contain fluoride in a high concentration ☐

Studies showed that areas with no fluoridation did not have higher levels of decay than areas that did fluoridate ☐

Examiner only

10. *(a)* Global warming is believed to be mainly the result of increasing levels of carbon dioxide in the atmosphere.

(i) State how the balance of carbon dioxide and oxygen is maintained in the atmosphere. Explain why levels of carbon dioxide are increasing and how this leads to global warming. [3]

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....

(ii) Carbon capture and storage (CCS) is one possible method of reducing global warming.

Describe briefly how CCS is carried out. [2]

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Examiner only

(b) Oxides of nitrogen are released in vehicle exhaust fumes. They can cause smog in busy city centres as well as acid rain.

(i) One oxide of nitrogen contains 30.4 g of nitrogen and 69.6 g of oxygen.

Find the simplest formula of this oxide. You **must** show your working. [3]

$A_r(\mathrm{N}) = 14$ $A_r(\mathrm{O}) = 16$

Simplest formula ....

(ii) The oxide with this simplest formula is found to have a relative molecular mass of 92. Find its molecular formula. [2]

Molecular formula ....

10

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2				Group							3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

 (3410UA0-1)

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3410UA0-1

Z22-3410UA0-1

FRIDAY, 17 JUNE 2022 – AFTERNOON

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**9**	
3.	**6**	
4.	**9**	
5.	**5**	
6.	**8**	
7.	**5**	
8.	**7**	
9.	**11**	
10.	**9**	
11.	**6**	
Total	**80**	

3410UA01
01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **9**(a) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

JUN223410UA0101

10. (a) Tables **A**, **B**, **C** and **D** show the results recorded by four different groups in a class investigating the reactivity of Group 7 elements (the halogens).

Each group added iodine, bromine and chlorine to solutions of sodium halides.

A tick (✓) indicates when a reaction took place and a cross (×) indicates no reaction. Only **one** table shows the expected results for this experiment.

Table **A**

	sodium chloride	sodium bromide	sodium iodide
bromine	✓	✓	×
chlorine	✓	×	✓
iodine	✓	✓	✓

Table **B**

	sodium chloride	sodium bromide	sodium iodide
bromine	✓	✓	✓
chlorine	×	×	✓
iodine	×	✓	✓

Table **C**

	sodium chloride	sodium bromide	sodium iodide
bromine	×	×	✓
chlorine	×	✓	✓
iodine	×	×	×

Table **D**

	sodium chloride	sodium bromide	sodium iodide
bromine	×	✓	×
chlorine	✓	×	✓
iodine	×	✓	✓

Examiner only

(i) Give the **letter** of the table which shows the expected results for this experiment. Explain why these are the expected results. [3]

Letter ....

....

....

....

(ii) Write a balanced symbol equation for the reaction between chlorine, Cl_2, and sodium iodide solution. [3]

....

(b) A sample of iron bromide with a mass of 37.0 g contains 7.0 g of iron. Find the simplest formula of the iron bromide.

You **must** show your working. [3]

A_r(Fe) = 56 A_r(Br) = 80

Simplest formula ....

9